

Combating viruses

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Viruses

- A computer virus is a self-replicating program that produces its code by attaching copies of itself to other executable code and operates without the knowledge or consent of the user.
- Like a biological virus, a computer virus is contagious and can contaminate other files; however, viruses can infect external machines only with the assistance of computer users.
- Some viruses affect computers as soon as their code is executed; other viruses remain dormant until a pre-determined logical circumstance is met.

Characteristics of Viruses

- The performance of a computer is affected by a virus infection.
- This infection can lead to data loss, system crash, and file corruption. Some of the characteristics of a virus are as follows:
  - Infects other programs
  - Transforms itself
  - Encrypts itself
  - Alters data
  - Corrupts files and programs
  - Replicates itself

Purpose of Creating Viruses

- An attacker creates a virus for the following purposes:
  - Inflict damage on competitors
  - Realize financial benefits
  - Vandalize intellectual property
  - Play pranks
  - Conduct research
  - Engage in cyber-terrorism
  - Distribute political messages
  - Damage network or computers
  - Gain remote access to the victim's computer

Indications of Virus Attack

- Indications of virus attacks arise from abnormal activities. Such activities reflect the nature of a virus by interrupting the regular flow of a process or a program
- Some indications of computer virus infection are as follows:
  - Processes require more resources and time, resulting in degraded performance
  - Computer beeps with no display
  - Drive label changes and OS does not load
  - Constant antivirus alerts
  - Computer freezes frequently or encounters an error such as BSOD
  - Files and folders are missing
  - Suspicious hard drive activity
  - Browser window “freezes”
  - Lack of storage space
  - Unwanted advertisements and pop-up windows

Stages of Virus Lifecycle

Design: Development of virus code using programming languages or construction kits.

Replication: The virus replicates for a period within the target system and then spreads itself.

Launch: The virus is activated when the user performs specific actions such as running an infected program.

Detection: The virus is identified as a threat infecting target system.

Incorporation: Antivirus software developers assimilate defenses against the virus.

How a virus works

- **Infection Phase**
  - **Programs** modified by a virus infection can enable virus functionalities to run on the system.
  - The virus infects the target system after it is triggered and becomes active upon the execution of infected programs, because the program code leads to the virus code. The two most important factors in the infection phase of a virus are as follows: o Method of infection o Method of spreading

How a virus works

Attack Phase

- Once viruses spread throughout the target system, they start corrupting the files and programs of the host system.
- Some viruses can trigger and corrupt the host system only after the triggering event is activated. Some viruses have bugs that replicate themselves and perform activities such as deleting files and increasing session time.
- Viruses corrupt their targets only after spreading as intended by their developers.

How a virus works

- In the attack phase:
  - Viruses execute upon triggering specific events
  - Some viruses execute and corrupt via built-in bug programs after being stored in the host's memory
  - The latest and most advanced viruses conceal their presence, attacking only after thoroughly spreading through the host

Types of viruses

- Viruses are categories according to their functioning and targets.
- Some of the most common types of computer viruses that adversely affect the security of systems are listed below:

I. System or Boot Sector Virus
II. File Virus
III. Polymorphic virus
IV. Email Virus
V. Stealth virus
VI. Network virus

System or Boot Sector Viruses

- A boot sector virus is a type of virus that infects the boot sector of floppy disks or the Master Boot Record (MBR) of hard disks (some infect the boot sector of the hard disk instead of the MBR).
- The infected code runs when the system is booted from an infected disk, but once loaded it will infect other floppy disks when accessed in the infected computer.
- While boot sector viruses infect at a BIOS level, they use DOS commands to spread to other floppy disks.

System or Boot Sector Viruses

SYMPTOMS

- A boot sector virus can cause a variety of boot or data retrieval problems. In some cases, data disappear from entire partitions.
- In other cases, the computer suddenly becomes unstable. Often the infected computer fails to start up or to find the hard drive.
- Also, error messages such as "Invalid system disk" may become prevalent.
- Examples of boot- sector viruses are **Michelangelo and Stoned**.

File viruses

- File viruses (or parasitic viruses) use arbitrary existing files as hosts.
- Usually the virus prepends the body of its code to the beginning of, or appends the body of its code to the end to the host file, in which case the original file contents remain intact, except that the OEP (original entry point) is modified, so that the virus code is executed before the original, legitimate code.
- File viruses are commonly found in executable files such as .exe, .vbs or a .com files. If you run an executable file that is infected with a file virus, it can potentially enter your computer's memory and subsequently run your computer.
- Two types of file viruses include direct action and resident viruses, which each have their own unique characteristics and methods for removal

File viruses

- A resident virus is **a kind of computer virus that hides and stores itself within the computer memory**, which then allows it to infect any file that is run by the computer, depending on the virus' programming.
  - a resident virus installs themselves on the computer and are therefore considered more dangerous and trickier to eradicate.
- A direct action computer virus is **a class of self-replicating malware that's attached to an executable file**. They are typically embedded in otherwise legitimate programs that require execution to run. After downloading and executing an infected program, the direct action computer virus will spread.
- File viruses hide their presence using stealth techniques to reside in a computer's memory in the same way as system sector viruses.
- They do not show any increase in file length while performing directory listing.

Polymorphic Viruses

- Such viruses infect a file with an encrypted copy of a polymorphic code already decoded by a decryption module.
- Polymorphic viruses modify their code for each replication to avoid detection.
- They accomplish this by changing the encryption module and the instruction sequence. Polymorphic mechanisms use random number generators in their implementation.
- A virus that mutates with every infection, making detection by the "signature" of the virus impossible.

Logic Bomb Viruses

- A logic bomb is a virus that is triggered by a response to an event, such as the launching of an application or when a specific date/time is reached, where it involves logic to execute the trigger.
- For example, cyber-criminals use spyware to covertly install a keylogger on your computer.
- The keylogger can capture keystrokes, such as usernames and passwords. The logic bomb is designed to wait until you visit a website that requires you to log in with your credentials, such as a banking site or social network.
- Consequently, the logic bomb will be triggered to execute the keylogger, capture your credentials, and send them to a remote attacker.
- When a logic bomb is programmed to execute on a specific date, it is referred to as a time bomb. Time bombs are usually programmed to set off when important dates are reached, such as Christmas and Valentine's Day.

Stealth viruses

- A stealth virus can disguise itself by using certain tactics to prevent being detected by antivirus software, uses various mechanisms to avoid detection by antivirus software.
- Stealth viruses hide in files, partitions and boot sectors and are adept at deliberately avoiding detection.
- These tactics include altering its file size, concealing itself in memory, and so on.
- It accomplishes this by temporarily moving itself away from the infected file and copying itself to another drive and replacing itself with a clean file.
- This type of virus is nothing new, in fact, the first computer virus, dubbed Brain, was a stealth virus.
- A good antivirus should be able to detect a stealth virus lurking on your hard drive by checking the areas the virus infected and evidence in memory.

NETWORK VIRUSES

- The strictest definition of a "network virus" describes a relatively new type of malware that spreads from computer to computer without having to drop a file-based copy of itself in any of the affected computers.
- A network virus makes use of networking protocols and/or applications to spread.
- These viruses exist only as network packets, when they move from one computer to another, and in memory.
- Network viruses make uses of system network mechanisms, search local and remote system information, monitor network traffic, take advantage of system and network vulnerabilities, and build network connections.

E-mail Viruses

- An e-mail virus refers to computer code sent to you as an e-mail attachment, which if activated, will result in some unexpected and usually harmful effects, such as destroying specific files on your hard disk and causing the attachment to be emailed to everyone in your address book.
- Email viruses perform a wide variety of activities, from creating pop-ups to crashing systems or stealing personal data.
- The first rapidly spreading e-mail viruses, such as Melissa, made use of a Microsoft Word macro embedded in an attachment. If the recipient opens the e-mail attachment, the Word macro is activated.
- Email is also one of the common methods of delivering viruses/malware to victims.

E-mail Viruses

- Email viruses can be packaged and presented in a variety of different ways. Some can easily be spotted as malicious by virtue of subject lines that don't make sense, suspicious sender or other header fields and body content that looks off in some way
- To avoid email virus attacks, you should never open (or double-click on) an e-mail attachment unless you know who sent it and what the attachment contains; in addition, you must install and use antivirus software to scan any attachment before you open it.

WHAT WE CAN DO TO PREVENT A VIRUS ATTACK

- Develop a Virus Protection Plan
- Install Network-Based Anti-Virus and Firewall
- Scan, Update, and Upgrade Automatically
- Back Up Your Data Regularly
- Consider Every Disk, Program, and Email Attachment as a Threat
- Use Caution When You Download Files from the Internet
- Be Aware of Virus Hoaxes
- Block executables file attachments
- Educate Your Network Users
- Avoid using accounts with administrative privileges on an infected network.
- Strong password on Shared Drives/folder
- Keep all applications (including your operating system) patched.

Virus elimination

- The best defense against a virus begins with awareness and ends with prevention.
- Prevention can be achieved through installation of antivirus software's
- Once antivirus software is installed in a computer system, it protects the computer system from such malicious software and ensures that any program that runs in the computer system is authorized.
- Backup all your critical data

THE STEPS IN THE VIRUS REMOVAL PROCESS

- Determine the type of virus
- Isolate all infected systems and hard disks.
- Make sure you have a clean hard disk drive formatted as a system disk
- Use the clean system disk to boot up all systems with suspected infection
- Scan every physical and logical hard disk, as well every USB Drives.
- Clean the infected server.

Computer Virus Removal- Step by Step Procedure

I. Safe Mode
II. Delete Temporary Files
III. Download Virus/Malware Scanner
IV. Run a Virus/Malware Scan
V. Reinstall the Software or Damaged Files

Computer Virus Removal- Step by Step Procedure

Safe Mode

- Restart your system and repeatedly press F8 for Advanced Boot Options menu and in that choose Safe Mode with Networking and press Enter.
- By doing so, it is easy to tackle nefarious files as they're not actually running or active. And, disconnect the internet at all costs, it prevents the infection from spreading.

Delete Temporary Files

- Delete temporary files before you start running a virus scan. It helps free up disk space, speed up the virus scanning process and even get rid of some virus or malware.
- Make use of the Disk Cleanup tool to delete the temporary files from the computer. Go to the start menu, select All Programs, click on Accessories, System Tools, and then click Disk Cleanup.

Computer Virus Removal- Step by Step Procedure

Download Virus/Malware Scanner

Running a scanner is not enough to remove harmful infection but it is sufficient for eliminating standard infections. So, download a virus/malware scanner to narrow down on the issue.

It is always better to go with a real-time anti-virus program since it keeps running in the background constantly checking for viruses and malware.

Please Note: Never install more than one real-time anti-virus program on a system as it creates problems.

Run a Virus/Malware Scan

Now that you have successfully downloaded and installed the Virus/Malware scanner, run the on-demand scanner first and thereafter run your real-time scanner.

One of them will effectively remove your computer virus or malware.

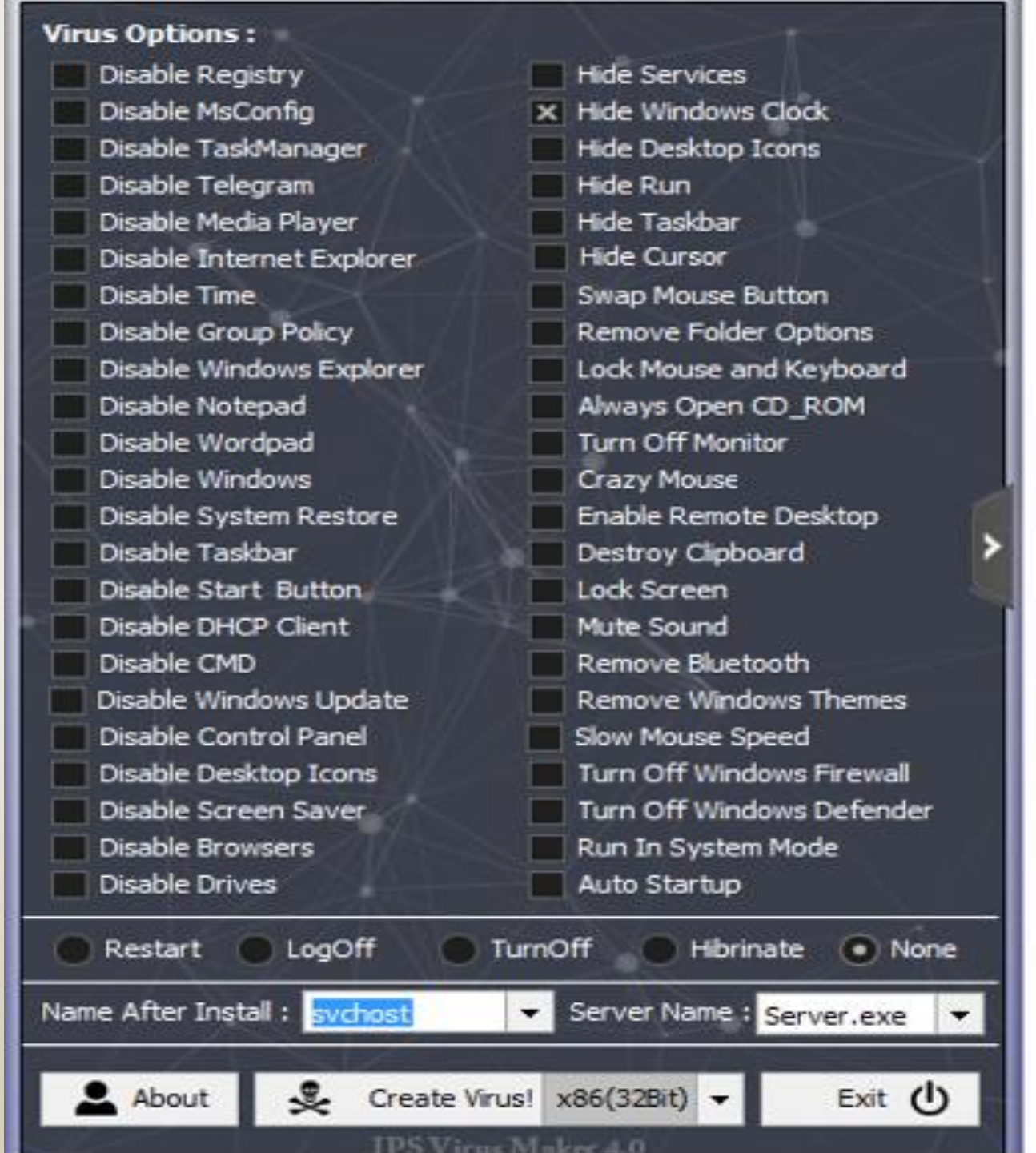
Computer Virus Removal- Step by Step Procedure

Reinstall the Software or Damaged Files

- After successful virus removal from the computer, you will have to reinstall the files and programs that were damaged by the virus or malware. Make use of the backups for re-installation. In simple, do the backups regularly and stay protected

Virus Making tools

- JPS virus maker
- DELmE's Batch Virus Generator
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Virus Options :
Disable Registry
Disable MsConfig
Disable TaskManager
Disable Telegram
Disable Media Player
Disable Internet Explorer
Disable Time
Disable Group Policy
Disable Windows Explorer
Disable Notepad
Disable Wordpad
Disable Windows
Disable System Restore
Disable Taskbar
Disable Start Button
Disable DHCP Client
Disable CMD
Disable Windows Update
Disable Control Panel
Disable Desktop Icons
Disable Screen Saver
Disable Browsers
Disable Drives
Hide Services
Hide Windows Clock
Hide Desktop Icons
Hide Run
Hide Taskbar
Hide Cursor
Swap Mouse Button
Remove Folder Options
Lock Mouse and Keyboard
Always Open CD_ROM
Turn Off Monitor
Crazy Mouse
Enable Remote Desktop
Destroy Clipboard
Lock Screen
Mute Sound
Remove Bluetooth
Remove Windows Themes
Slow Mouse Speed
Turn Off Windows Firewall
Turn Off Windows Defender
Run In System Mode
Auto Startup
Restart
LogOff
TurnOff
Hibrinate
None
Name After Install :
svchost
Server Name :
Server.exe
About
Create Virus!
x86(32Bit)
Exit

Additional resources

- https://mycheckanswer.wordpress.com/my-knowledge/about/jps-virus-maker-3-0/
- https://www.smithsonianmag.com/science-nature/top-ten-most-destructive-computer-viruses-159542266/